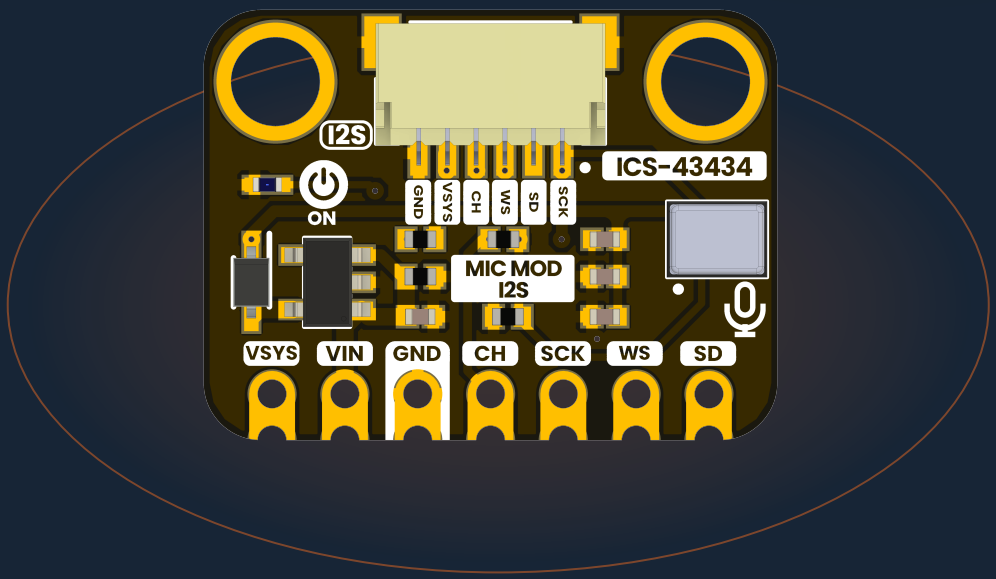
PRODUCT REFERENCE MANUAL

DEVLAB DIGITAL AUDIO MODULE

DevLab I2S ICS-43434 MEMS Microphone

MPN: UE0149

Version 0.1.0



Compact ICS-43434 MEMS microphone module with 24-bit I²S audio output, selectable channel assignment, and onboard 3.3 V regulation.

I²S AUDIO | 24-BIT OUTPUT | CHANNEL SELECT | MEMS

DOCUMENT MAP

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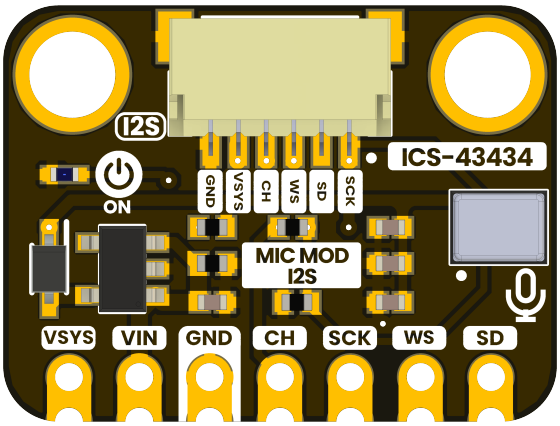
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Description

The UNIT DevLab I²S ICS-43434 MEMS Microphone is a compact digital-audio module based on the bottom-port ICS-43434 MEMS microphone. The microphone integrates signal conditioning, a 24-bit analog-to-digital converter, decimation and anti-aliasing filters, power management, and an I²S output.



Applications

- Voice capture and recognition prototypes
- Intercom and teleconferencing systems
- Acoustic sensing and security projects
- Stereo or microphone-array experiments
- Educational I²S audio acquisition

Hardware Features

- ICS-43434 bottom-port digital MEMS microphone
- 24-bit I²S serial audio output
- Left/right time-slot selection through the CH signal
- AP2112K fixed 3.3 V onboard regulator
- Seven 2.54 mm edge connections
- Six-position, 1.0 mm-pitch JST-SH connector
- Power indicator and selectable solder options shown on the V1.0 artwork

1 The Module

1.1 Board Identification

Item	Value
Product	DevLab: I2S ICS-43434 MEMS Microphone
Manufacturer part number	UE0149
Hardware revision shown	V1.0
Microphone	ICS-43434
Digital audio interface	I^2S
Product-reference revision	0.1.0, preliminary

1.2 Package Contents

The BOM includes one four-conductor, 1.0 mm-pitch QWIIC-style harness. Confirm the contents of a purchased product with the distributor because package contents have not yet been formally released. The BOM also identifies J1 as a six-position connector, so harness-to-J1 compatibility remains to be verified.

2 Electrical Characteristics

2.1 Currently Documented Values

Symbol / parameter	Description	Value	Scope
VDD	ICS-43434 microphone supply	1.62 to 3.63 V	Microphone pin
ADC output	Digital sample width	24 bit	Microphone I^2S output
SNR	Signal-to-noise ratio	64 dBA typical	Microphone characteristic
Frequency response	Stated audio band	60 Hz to 20 kHz	EMMIC-ICS43434 reference
Sensitivity tolerance	Unit-to-unit tolerance	±1 dB	Microphone characteristic
VREG	U3 regulated output	3.3 V nominal	BOM identification

The allowed voltage at the module's VIN pin and the electrical behavior of VSYS have not been verified from a released schematic. Do not substitute the microphone VDD range or the AP2112K component rating for a module input rating.

2.2 Electrical Precautions

- Connect GND between the host and module before applying power or clocks.
- Drive SCK, WS, and CH only within the validated logic domain; the module-level input thresholds remain pending.
- Do not drive SD; it is an output from the microphone.
- Keep the bottom acoustic port free from debris, flux, adhesive, and enclosure pressure.
- Confirm VIN, VSYS, and JST supply behavior on the schematic or physical board before first power-up.

3 Functional Overview

3.1 Audio Signal Path

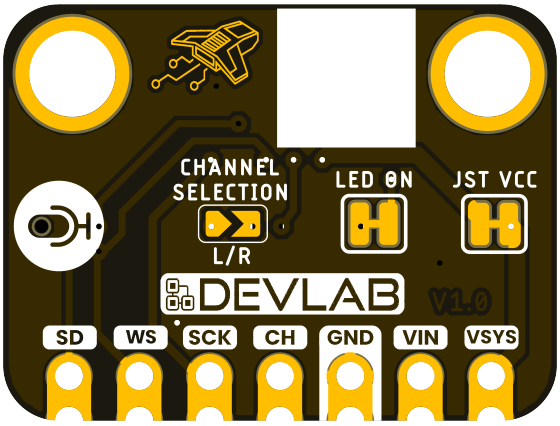
The ICS-43434 senses pressure at its bottom acoustic port, conditions the MEMS signal, converts it to digital audio, and places 24-bit samples on SD. A host controller supplies SCK and WS; no master clock line is exposed by the module.

3.2 Channel Selection

CH selects the I²S time slot used by the microphone. A low level selects the left channel and a high level selects the right channel. Two modules can share the clock and data timing when their channel selections and data connection are implemented according to the ICS-43434 application requirements.

3.3 Power Section

The manufacturing BOM identifies an AP2112K-3.3TRG1 regulator, Schottky diode, decoupling capacitors, orange indicator LED, and associated resistors. The exact VIN/VSYS power path and solder-option defaults remain pending until a released module schematic is available.



4 Connectors and Pinout

4.1 Edge Connections

Board label	Direction	Function
VSYS	Power	Module system rail
VIN	Input	Module supply input; allowed range pending
GND	Power	Common ground
CH	Input	Left/right I^2S channel selection
SCK	Input	I^2S serial bit clock
WS	Input	I^2S word select / left-right clock
SD	Output	I^2S serial audio data

4.2 Six-Position JST-SH Connector

The V1.0 top artwork labels the connector signals, from the leftmost contact in the rendered top view, as GND, VSYS, CH, WS, SD, and SCK. Verify orientation on the physical board before making a cable; the present document does not assign connector pin numbers.

4.3 Solder Options

The bottom artwork identifies L/R, LED ON, and JST VCC options. Their default copper states and exact circuit connections are pending schematic verification.

5 Preliminary Operation Notes

The firmware under `software/examples/i2s/` is experimental and remains under test, so it is intentionally not reproduced as a validated example here.

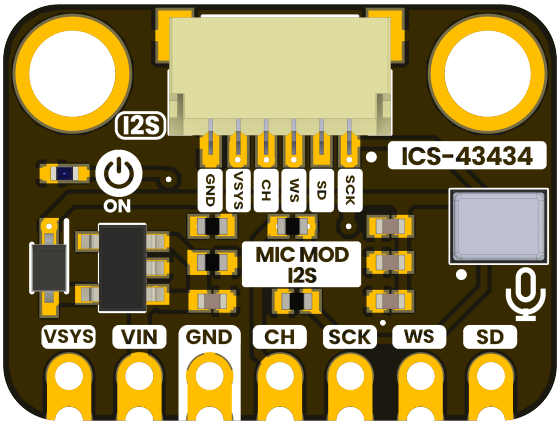
For initial bench work:

1. Inspect the board and ensure the bottom acoustic port is unobstructed.
2. Verify the `VIN`/`VSYS` power path against a schematic or measured board.
3. Connect the host and module grounds.
4. Connect host I²S bit clock to `SCK`, word select to `WS`, and receive data to `SD`.
5. Set `CH` low for the left time slot or high for the right time slot.
6. Start with conservative audio settings and confirm framing using a logic analyzer before relying on captured sample values.

Host GPIO numbers are controller-specific and are not module pin numbers. Clock limits, tested sample rates, sample alignment, and controller examples will be added after validation.

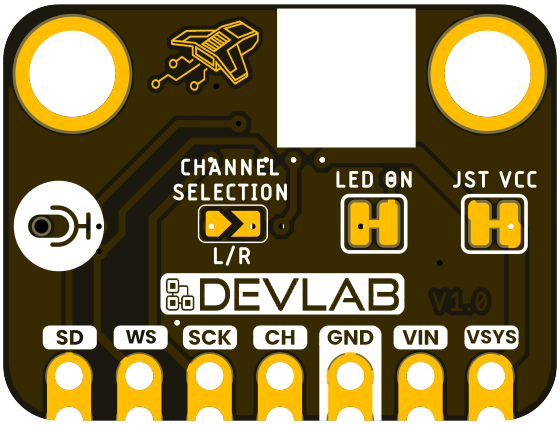
6 Mechanical Information

6.1 Top View



UE0149 I2S microphone top view.

6.2 Bottom View



UE0149 I2S microphone bottom view.

Keep the bottom acoustic port clear of adhesive, contamination, and mounting hardware.

6.3 Board Dimensions

The dimension drawing is stored with the board images.

Use the labeled drawing for mechanical layout. Verify tolerances and connector clearance on the physical assembly before designing a carrier or enclosure.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	https://uelectronics.com/
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

Product specifications and documentation may be revised. Use the repository and product page listed in the next chapter for the latest released material.

8 Reference Documentation

Reference	Location
Repository	UNIT-Electronics-MX/unit_devlab_i2s_ics_43434_mems_microphone
Hardware notes	Hardware README
Pinout PDFs	English · Spanish
Pinout images	English PNG · Spanish PNG
Microphone reference	EMMIC-ICS43434 PDF
Manufacturing BOM	ICS-43434 module BOM
Experimental firmware	I²S example
Top view	3D board view
Bottom view	3D board view
CAD model	STEP

The three-page EMMIC-ICS43434 file is a third-party breakout-board reference, not the complete manufacturer datasheet for the ICS-43434 and not a DevLab module datasheet. Use it only for the characteristics explicitly cited in this preliminary document.

9 Appendix

9.1 Source Hierarchy

This preliminary product reference follows the V1.0 board artwork for visible signal labels, the manufacturing BOM for fitted component identities, and the local EMMIC-ICS43434 document for the limited microphone values quoted here. A future released schematic and complete manufacturer datasheet take precedence if they conflict with this edition.

9.2 Items Pending Validation

- Module schematic and controlled pin-number drawing
- Module VIN and VSYS operating and absolute-maximum ratings
- Digital input thresholds, current consumption, and clock limits
- Tested sample rates and controller-specific I^2S configurations
- Board dimensions and mounting-hole coordinates
- Solder-option defaults and JST cable orientation
- Acoustic and environmental module-level characterization

9.3 Document Control

Field	Value
Product	DevLab: I2S ICS-43434 MEMS Microphone
Manufacturer part number	UE0149
Hardware artwork covered	V1.0
Product-reference revision	0.1.0, preliminary
Publication date	2026-08-03
Firmware status	Experimental; not validated by this edition